

Name: _____ Student ID number: _____

FINAL

No books or notes are allowed. **100 Points + (bonus, 15 Points), 180 Minutes.**

1.(10 points) Let f be an elliptic function with respect to the lattice $L = [\omega_1, \omega_2]$. Let P be a fundamental parallelogram. Prove that $\sum_{z \in P} \text{ord}_z(f)z \in L$. (Hint: apply the residue formula to the function $zf'(z)/f(z)$.)

2.(10 points) Consider the spaces $\mathbb{H}$, $\mathbb{C}$, $\mathbb{C} \setminus \mathbb{R}$, $\mathbb{C} \setminus \mathbb{R}_+$ and $\{z \in \mathbb{C} \mid -1 < \text{Re}(z) < 1\}$. Group them conformally and justify your answer.

3.(1) (5 points) Prove that any conformal map from $\mathbb{H}$ to $\mathbb{D}$ takes the form

$$e^{i\theta} \frac{z - \alpha}{z - \bar{\alpha}},$$

where $\theta \in \mathbb{R}, \alpha \in \mathbb{H}$.

(2) (5 points) Are $\mathbb{D}$ and $\mathbb{C} \setminus \{(-\infty, \frac{1}{4}]\}$ conformal? If so, find an explicit map and justify your answer; if not, also justify your answer.

4.(10 points) Evaluate the integral

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^4} dx.$$

(You are not required to simplify the result.)

5.(10 points) Verify that $\sin z$ has growth order 1, and use the Hadamard factorisation theorem to show the identity

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

6.(10 points) Let $\Omega \subset \mathbb{C}$ be a bounded open set, $\bar{\Omega}$ its closure. Let $f : \bar{\Omega} \rightarrow \mathbb{C}$ be a continuous function, which is holomorphic in Ω . Assume that f is not constant, and $|f(z)| = 1$ for any z on the boundary of Ω . Prove that $f(\Omega) = \mathbb{D}$. (Hint: you may apply the maximum principle and $\text{Aut}(\mathbb{D})$).

7.(1) (6 points) Let $g : \{z \in \mathbb{C} \mid |z| > 1\} \rightarrow \mathbb{C}$ be an injective holomorphic function with a convergent expansion

$$g(z) = z + d_0 + \frac{d_1}{z} + \frac{d_2}{z^2} + \dots,$$

show that $\sum_{n=0}^{\infty} n|d_n|^2 \leq 1$. (Hint: assume $r > 1$, use the fact that g maps the circle $\{|z| = r\}$ to a closed curve which encloses a set Ω with positive area, and apply the Green formula.)

(2) (10 points) (I) Let $f : \mathbb{D} \rightarrow \mathbb{C}$ be an odd injective holomorphic function with the expansion

$$f(z) = z + b_3 z^3 + b_5 z^5 + \dots,$$

prove that $|b_3| \leq 1$ with equality holds if and only if

$$f(z) = \frac{z}{1 + e^{i\theta} z^2}$$

for some $\theta \in \mathbb{R}$. (II) Let $h : \mathbb{D} \rightarrow \mathbb{C}$ be an injective holomorphic function with the expansion

$$h(z) = z + a_2 z^2 + a_3 z^3 + a_4 z^4 + \dots,$$

prove that $|a_2| \leq 2$ with equality holds if and only if

$$h(z) = \frac{z}{(1 + e^{i\theta} z)^2}$$

for some $\theta \in \mathbb{R}$. (Hint: for (I), apply (1) to the function g defined by $g(z) = 1/f(\frac{1}{z})$; for (II), apply part (I) to the function $z\sqrt{h(z^2)/z^2}$.)

(3) (4 points) Let $f : \mathbb{D} \rightarrow \mathbb{C}$ be an injective holomorphic function, show that $|f''(0)| \leq 4|f'(0)|$. For such f , give a characterization of those f such that the equality holds.

8. Let k be a fixed non-negative integer, f a non-constant meromorphic function on $\mathbb{H}$ satisfying that

$$(1) \quad f\left(\frac{az+b}{cz+d}\right) = (cz+d)^{2k} f(z) \text{ for any } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}).$$

(1) (5 points) Let $g(q_z) := f(z)$, where $q_z = e^{2\pi iz}$. Show that as a function of q , g is well-defined and meromorphic on the punctured disc $\mathbb{D} \setminus \{0\}$.

The function f is called meromorphic at ∞ if g extends to a meromorphic function on $\mathbb{D}$. If assume additionally that f is meromorphic at ∞ , then by (1) we can define $\text{ord}_\infty(f)$ by using $\text{ord}_0(g)$.

(2) (15 points) Assume additionally that f is meromorphic at ∞ . Let $\rho = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Prove that

$$\text{ord}_\infty(f) + \frac{1}{3} \text{ord}_\rho(f) + \frac{1}{2} \text{ord}_i(f) + \sum_{z \in \mathbb{H}/SL_2(\mathbb{Z}), z \neq i, \rho} \text{ord}_z(f) = \frac{k}{6},$$

where the sum is taken over points of $\mathbb{H}/SL_2(\mathbb{Z})$ which are not in the orbit of i, ρ . For $\mathbb{H}/SL_2(\mathbb{Z})$ (or the action of $SL_2(\mathbb{Z})$ on $\mathbb{H}$), see the appendix below. (Hint: let $P = \{z \in \mathbb{C} \mid \text{Im}(z) > 0, |z| \geq 1, |\text{Re}(z)| \leq \frac{1}{2}\}$, for t large enough, let $P_t = P \cap \{\text{Im}(z) \leq t\}$, integrate f'/f along the boundary of P_t , but modified by small arcs around the possible poles on the boundary.)

(3) (**bonus**, +15 points) Given a non-negative integer k , let A_k be the function space given by all such f that satisfies the equation (1), and is holomorphic in $\mathbb{H}$ and holomorphic at ∞ (i.e., g is holomorphic near 0.). Find A_k for all $k \leq 5$. (Hint: you may consider certain functions given by functions of lattices satisfying the equation (1).)

Appendix

Facts, definitions, notations and conventions that you may need in this final exam.

- $\mathbb{D} = \{|z| < 1\}$ is the unit disc; $\mathbb{H} = \{\text{Im}(z) > 0\}$ is the upper half plane; $\mathbb{R}_+ = \{(x, 0) \mid x \geq 0\}$.
- $SL_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, ad - bc = 1 \right\}$, similarly for $SL_2(\mathbb{R})$.
- A holomorphic map $f : U \rightarrow V$ is called conformal, if the inverse map $f^{-1} : V \rightarrow U$ exists and is also holomorphic. In this case, we call that U, V are conformally equivalent.
- Residue formula: $\int_C f dz = 2\pi i \sum_z \text{Res}_z(f)$.
- Let f be a meromorphic function, we use the convention that $\text{ord}_z(f) > 0$ if z is a zero, $\text{ord}_z(f) < 0$ if z is a pole, otherwise $\text{ord}_z(f) = 0$.
- Maximum principle: a non-constant holomorphic function defined in a connected open set cannot obtain its maximum in the interior.
- Any element of $\text{Aut}(\mathbb{D})$ takes the form $e^{i\theta} \frac{z-a}{1-\bar{a}z}$, where $\theta \in \mathbb{R}, a \in \mathbb{D}$.
- Any element of $\text{Aut}(\mathbb{H})$ takes the form $\frac{az+b}{cz+d}$, where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{R})$.
- Hadamard factorisation theorem: Let f be an entire function of growth order s , let $k = \lfloor s \rfloor$, let $a_1, a_2, \dots$ be the (non-zero) zeros of f , then $f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n)$, where P is a polynomial of degree $\leq k$, m is the order at 0, and $E_k(z) = (1-z)e^{z+z^2/2+\dots+z^k/k}$.
- Green formula: $\int_{\Omega} \left(\frac{\partial P}{\partial x} - \frac{\partial Q}{\partial y} \right) dx dy = \int_{\partial\Omega} P dy + Q dx$.

- The group $SL_2(\mathbb{Z})$ has an action on $\mathbb{H}$: for $\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$, $z \in \mathbb{H}$, $\alpha(z) = \frac{az+b}{cz+d}$.

The action has two generators:

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The set $P = \{z \in \mathbb{C} \mid \text{Im}(z) > 0, |z| \geq 1, |\text{Re}(z)| \leq \frac{1}{2}\}$ is a fundamental domain for $SL_2(\mathbb{Z})$, i.e., every orbit of $SL_2(\mathbb{Z})$ has one element in P , and two elements of P are in the same orbit if and only if they are on the boundary of P . Moreover, two points $z, z' \in P$ are in the same orbit if and only if: either they are on the vertical lines of P and $z = z' \pm 1$, or they are on the base arc of P and $zz' = -1$. In particular, this realizes the quotient space $\mathbb{H}/SL_2(\mathbb{Z})$ as the union of the interior of P and left half (with respect to the y -axis) boundary of P .